Name: ______________________

Starter quiz

Finding a length in composite shapes

1 Find the positive value of x that is a solution to the equation: $x^2 + 4 = 85$. $x =$ ______________________ . Fill in the blank

2 Select the fully factorised form of this expression: $42r + 56$. Tick **1** correct answer

- [ ] $2(21r + 28)$
- [ ] $7(6r + 8)$
- [ ] $14(7r)$
- [ ] $14(3r + 4)$
- [ ] $9(3r + 4)$

3 Find the positive value of x that is a solution to the equation: $3\times 7\times 2x^2 = 3\times 56$. $x =$ ______________________ Fill in the blank

4 Select the fully factorised form of this expression: $2kx^2 + 3kx$. Tick **1** correct answer

- [ ] $k(2x^2 + 3x)$
- [ ] $x(2kx + 3k)$
- [ ] $kx(2x + 3)$
- [ ] $kx(2x + 3k)$

5 A circle has a radius of 1.2 cm. Find the area of this circle. Give your answer in cm, rounded to 2 decimal places. Area = ______________________ cm². Fill in the blank

6 The area of a circle is 908 cm². Use the formula: $\text{area} = \pi\times r^2$ to form an equation and solve this equation to find the radius of the circle, rounded to the nearest integer. ______________________ cm. Fill in the blank